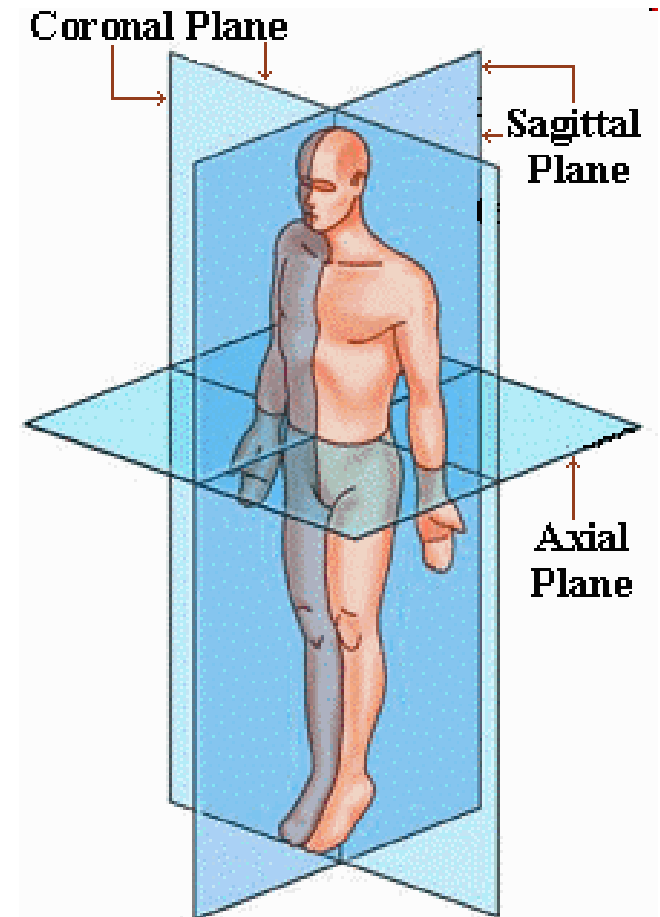
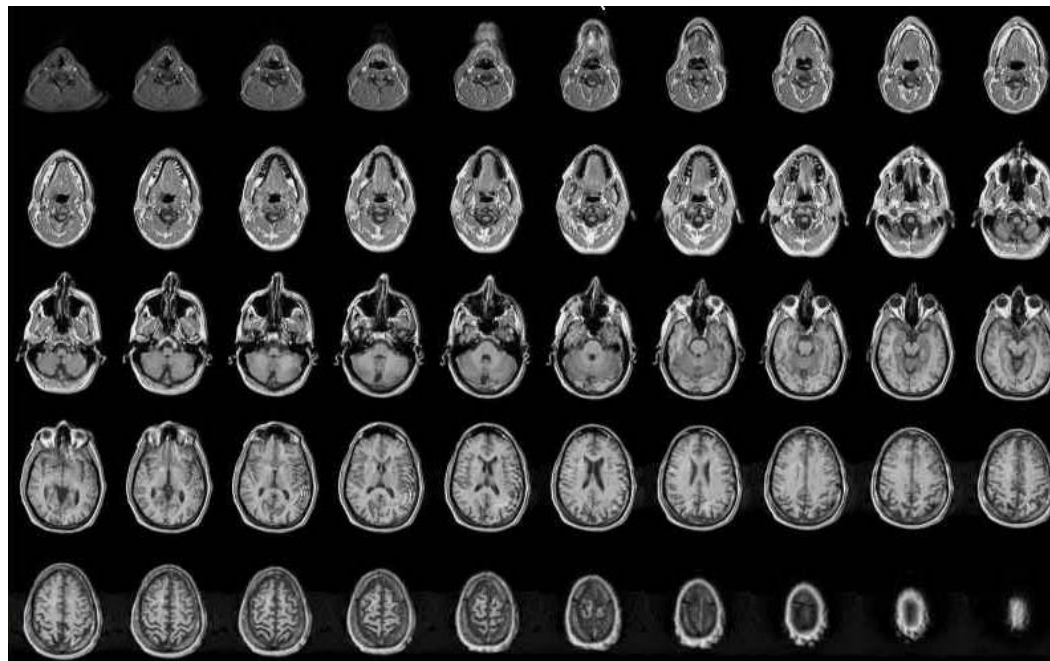


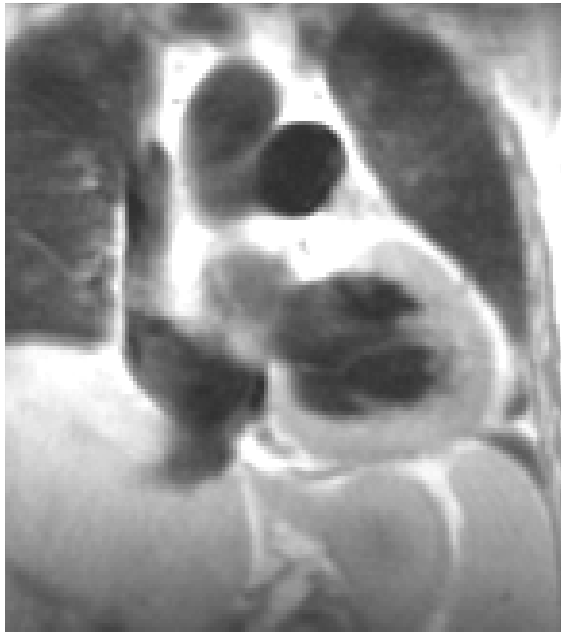
MRI – Image Reconstruction

Sample MRI: Sequence of 2D slices

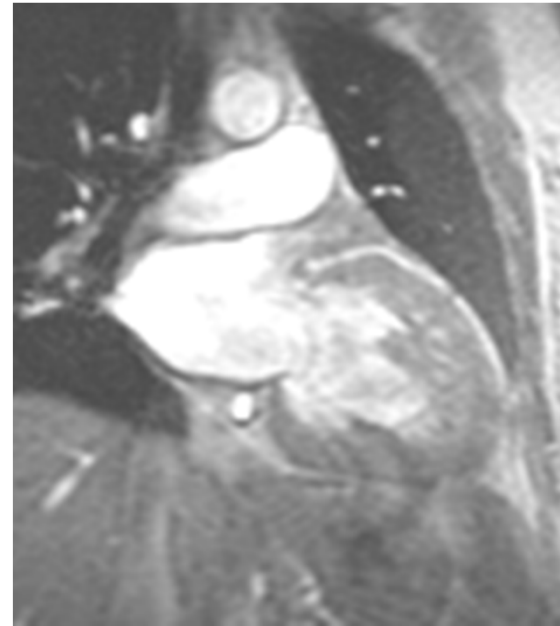


MRI: Physics meets Clinic (Part I)

Pulse Sequences: Spin Echo v/s Gradient Echo

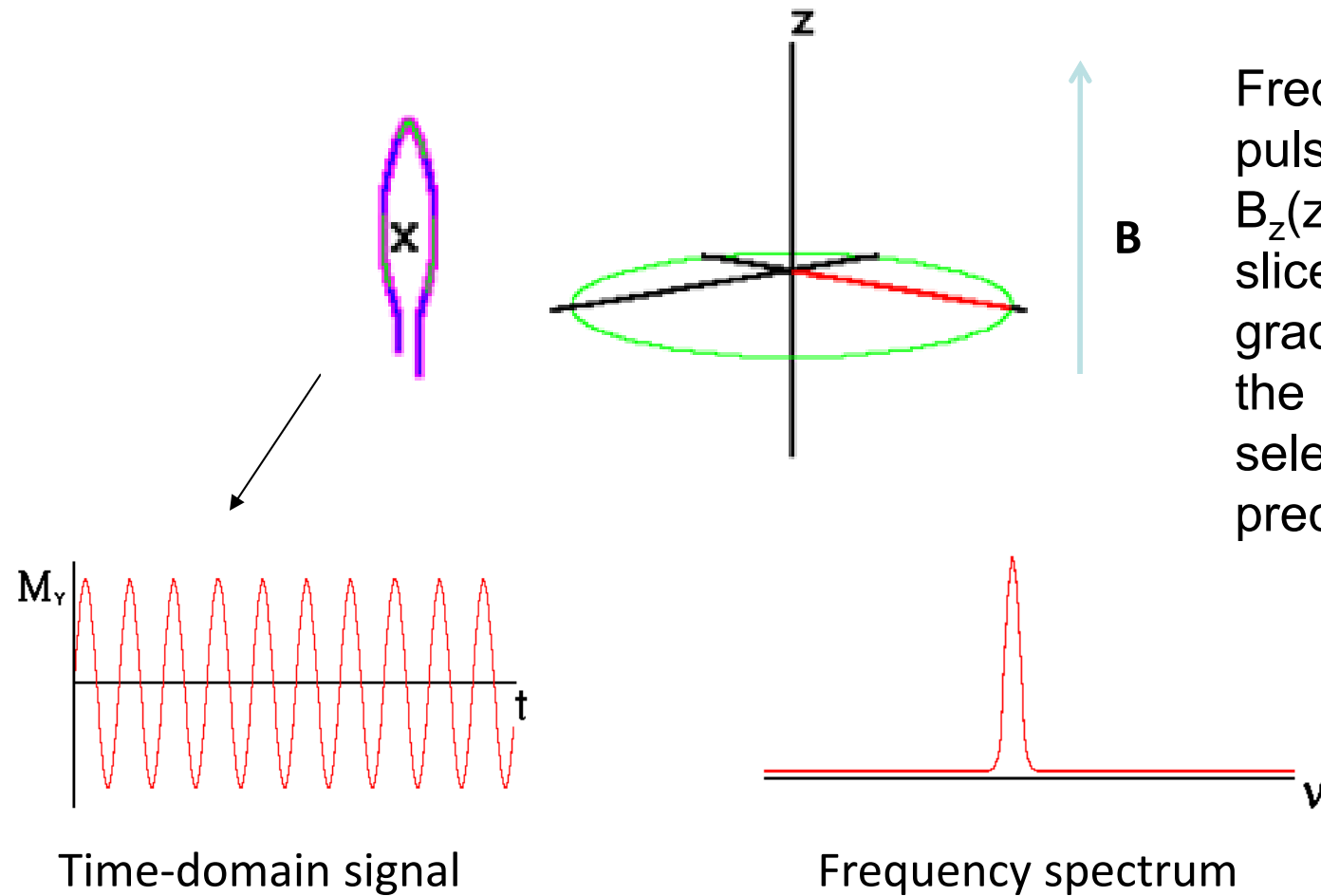


- Spin Echo

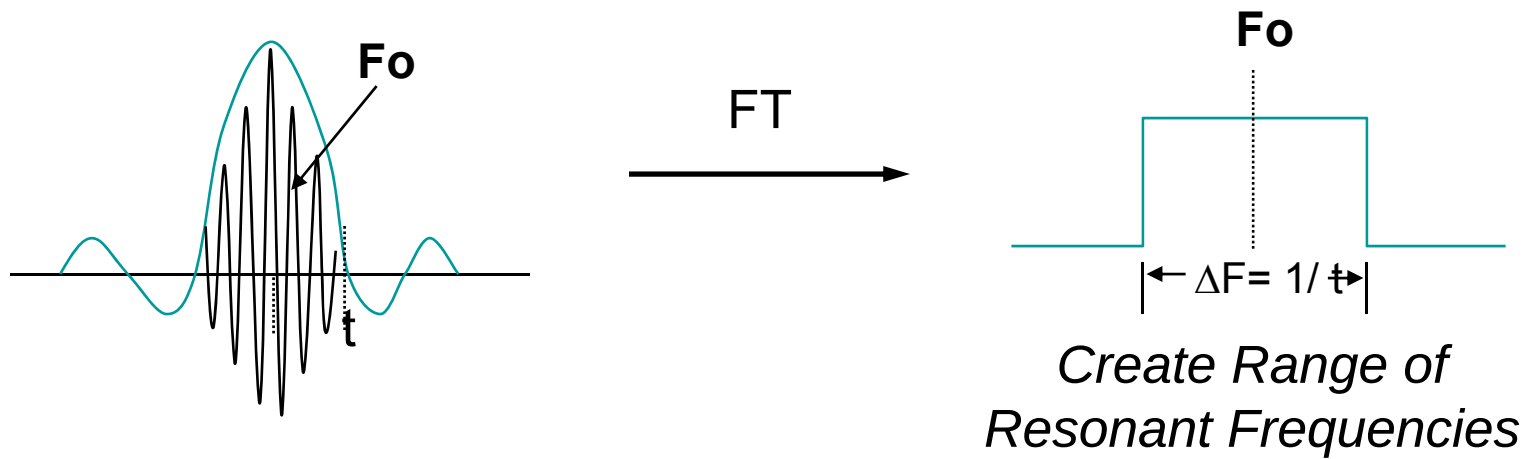
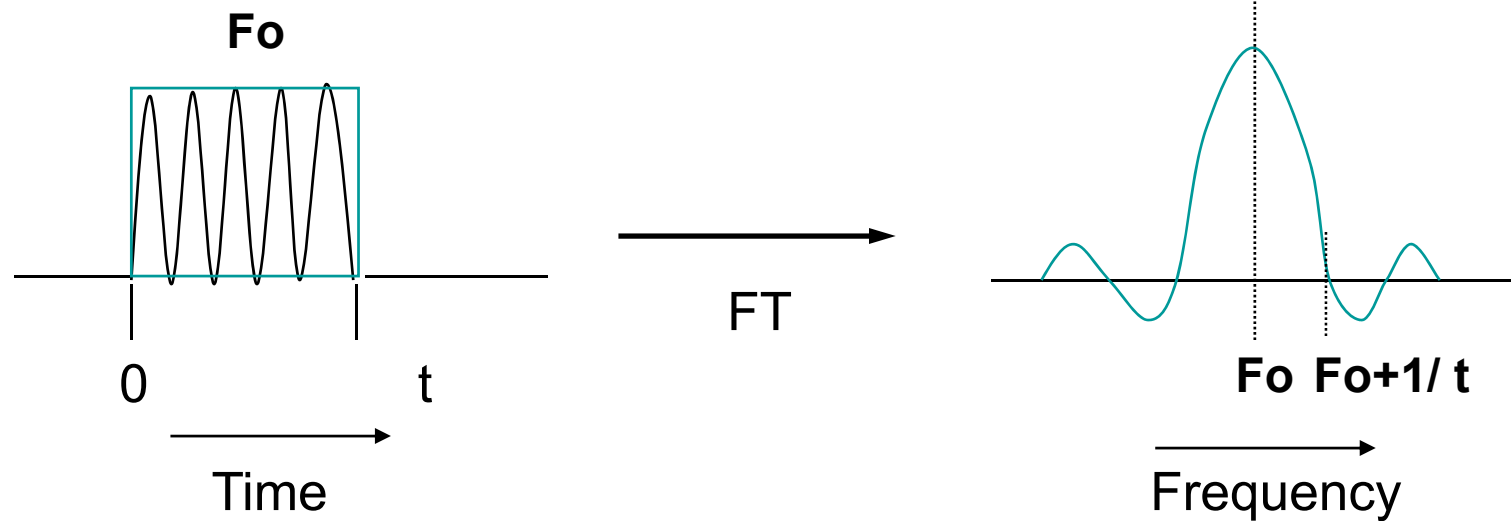


Gradient Echo

MR signal

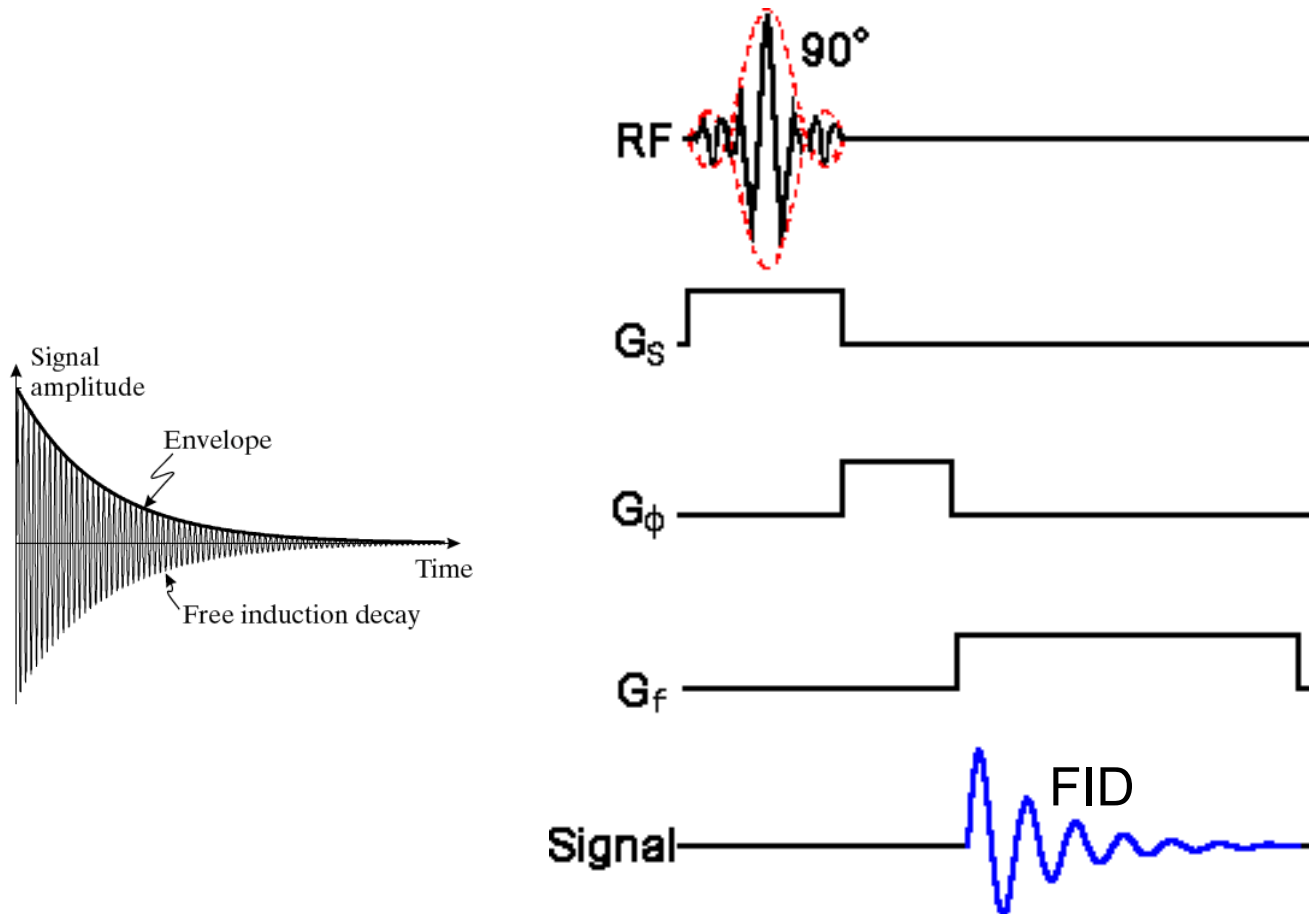


Electromagnetic Excitation Pulse (RF Pulse)

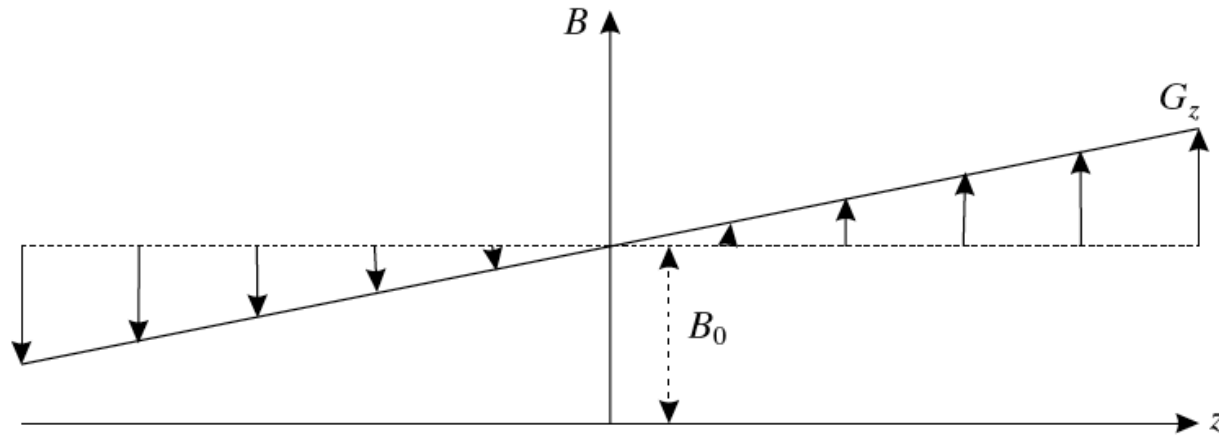


MRI Pulse Sequence

Most basic : free induction decay (FID)



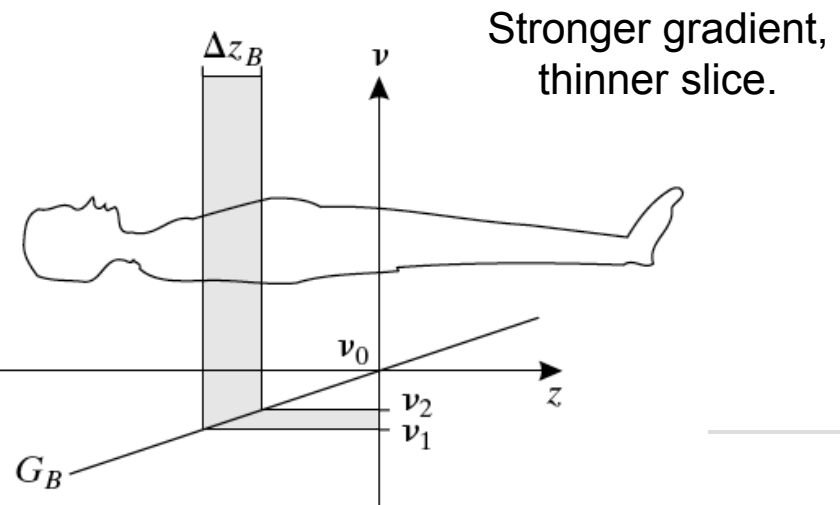
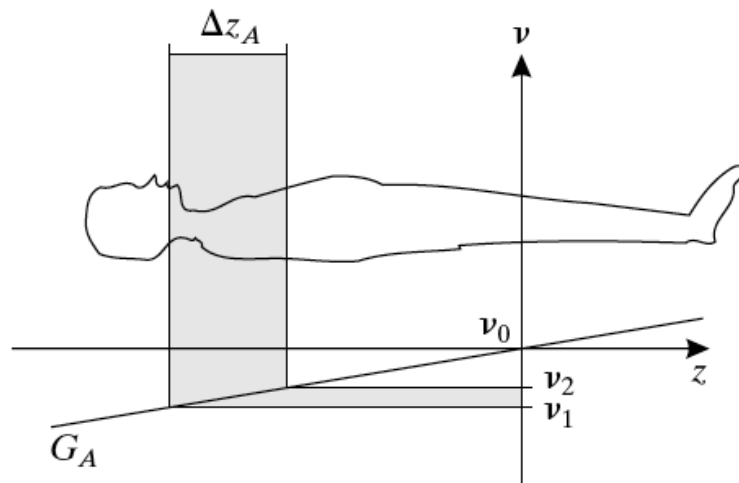
- Thickness and location of slice determined by strength of gradient and particular frequency



$$\bar{\nu} = \frac{\nu_1 + \nu_2}{2}$$

$$\Delta\nu = |\nu_2 - \nu_1|$$

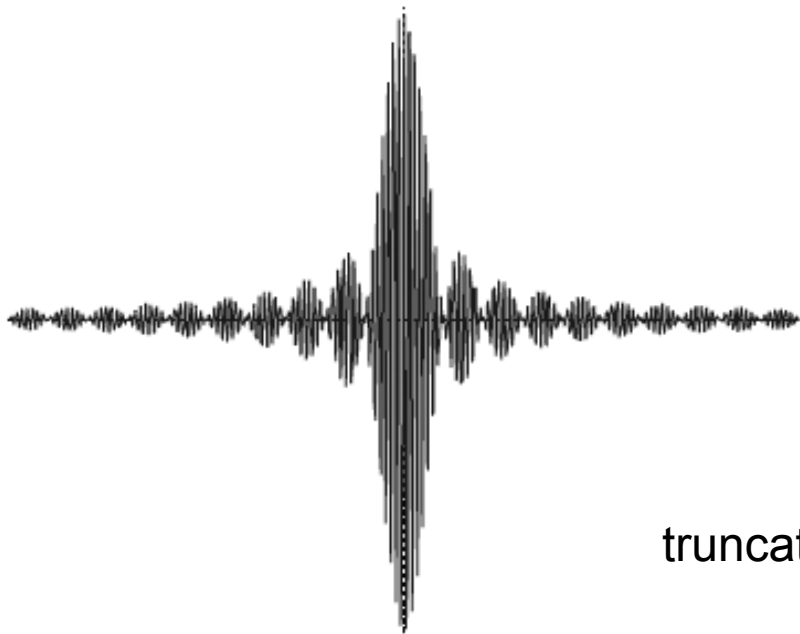
→ → → → →
Size and orientation of main field



Stronger gradient,
thinner slice.

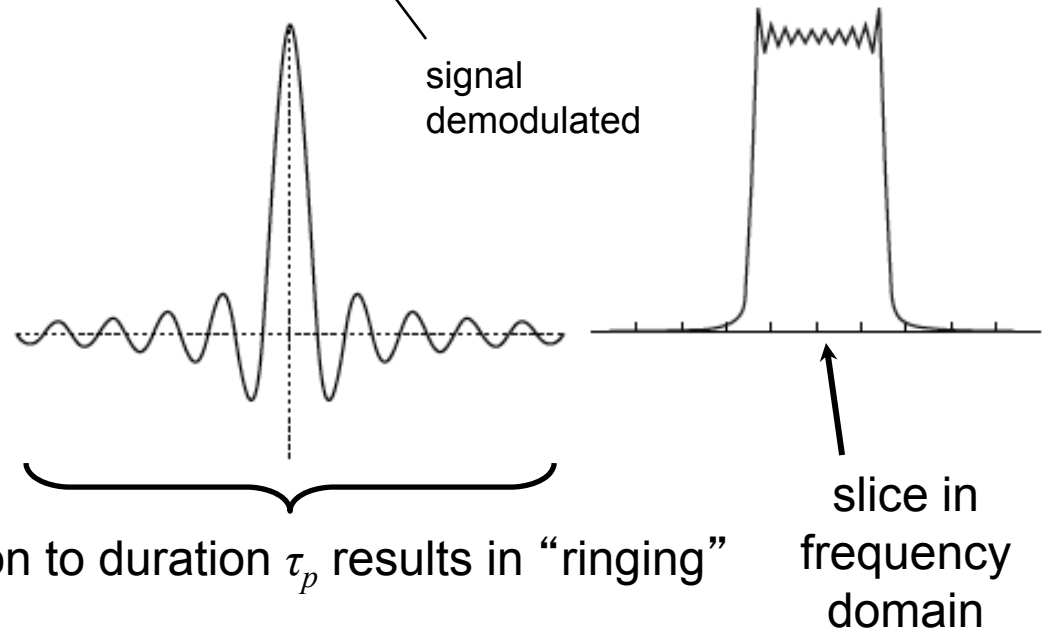
RF pulse,

$$s(t) = A\Delta\nu \operatorname{sinc}(\Delta\nu t) e^{j2\pi\bar{\nu}t}$$



its envelope and Fourier transform

$$B_1^e(t) = s(t)e^{-j2\pi\bar{\nu}t} \quad S(\nu) = A \operatorname{rect}\left(\frac{\nu - \bar{\nu}}{\Delta\nu}\right)$$



The resulting tip angle,

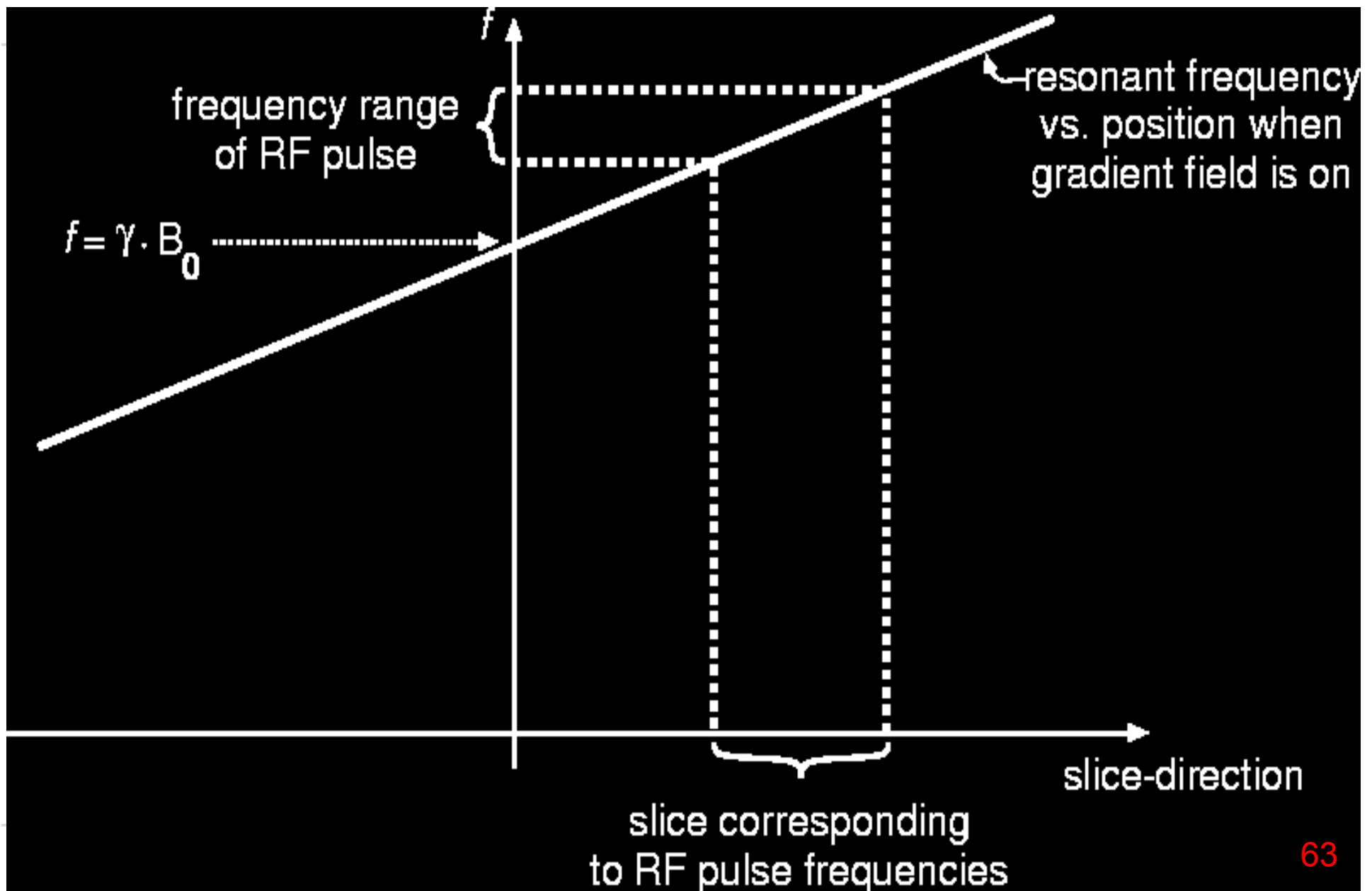
$$\alpha = \gamma \int_0^{\tau_p} B_1^e(t) dt$$

whose Fourier transform is...

$$\alpha(z) = \gamma A \operatorname{rect}\left(\frac{z - \bar{z}}{\Delta z}\right)$$

...constant across a slice

Slice Selection



Determining Slice Thickness

Resonance frequency range as the result of slice-selective gradient:

$$\Delta F = \gamma_H * G_{sl} * d_{sl}$$

The bandwidth of the RF excitation pulse:

$$\Delta\omega/2\pi$$

Thus the slice thickness can be derived as

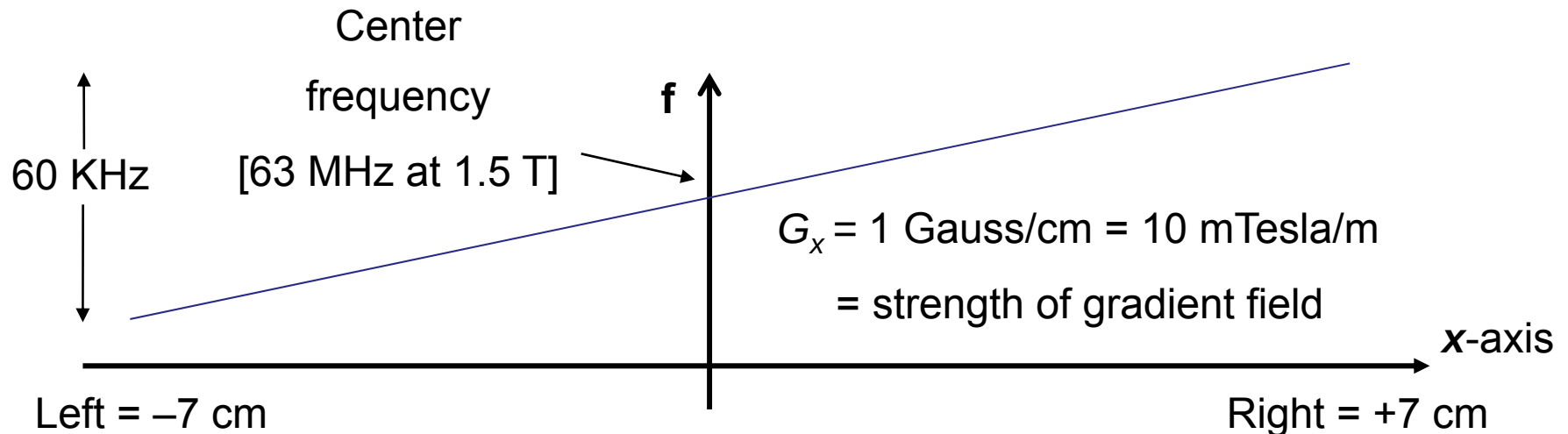
$$d_{sl} = \Delta\omega / (\gamma_H * G_{sl} * 2\pi)$$

Steps in 3D Localization

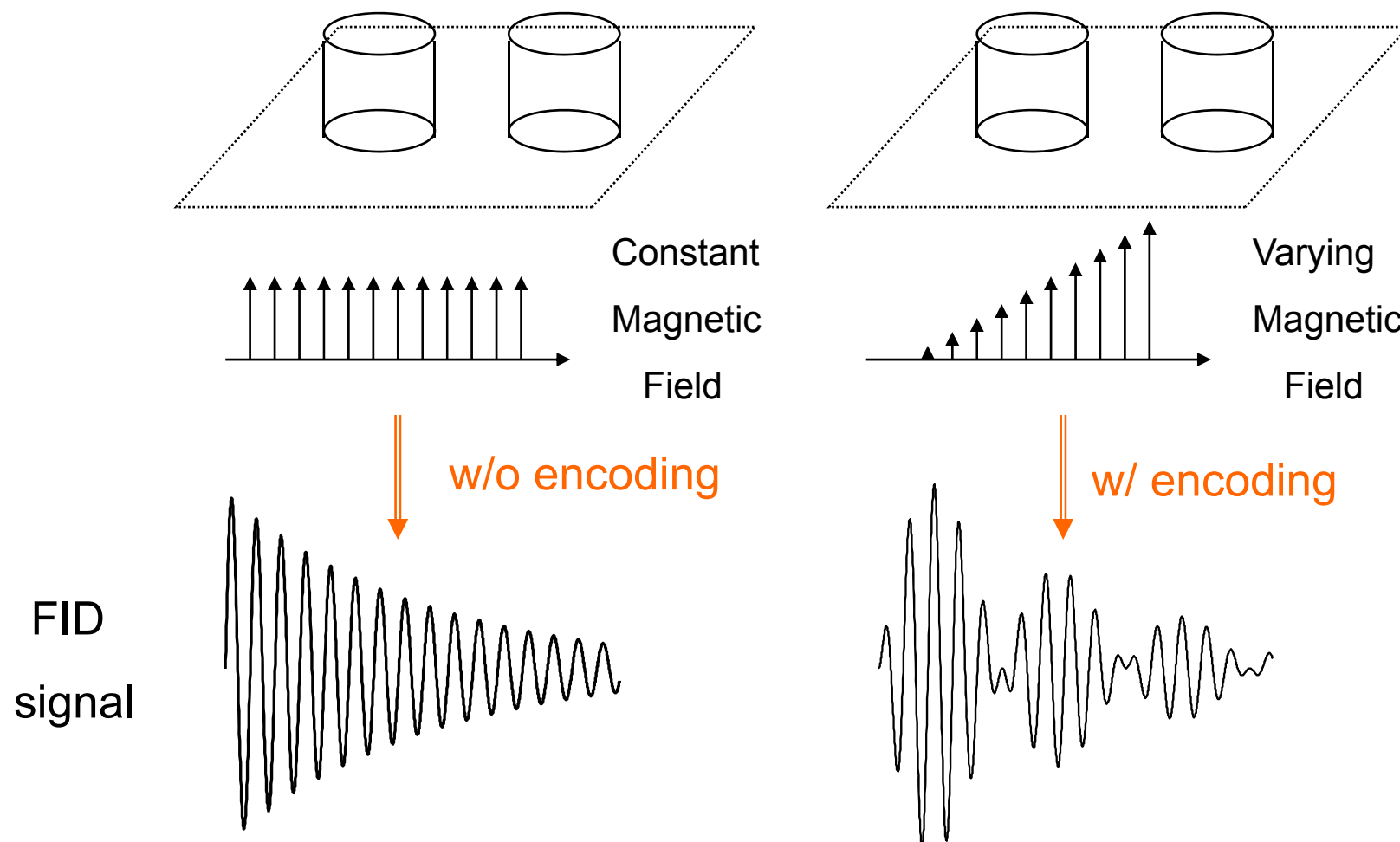
- Can only detect total RF signal from inside the form factor of the “RF coil” (the detecting antenna)
- ❶ Excite and receive M_{xy} in a thin (2D) slice of the subject
 - ➔ The RF signal we detect must come from this slice
 - ➔ Reduce dimension from 3D down to 2D
- ❷ Deliberately make magnetic field strength B depend on location within slice
 - ➔ Frequency of RF signal will depend on where it comes from
 - ➔ Breaking total signal into frequency components will provide more localization information
- ❸ Make RF signal phase depend on location within slice.

Gradient Fields: Spatially Nonuniform B :

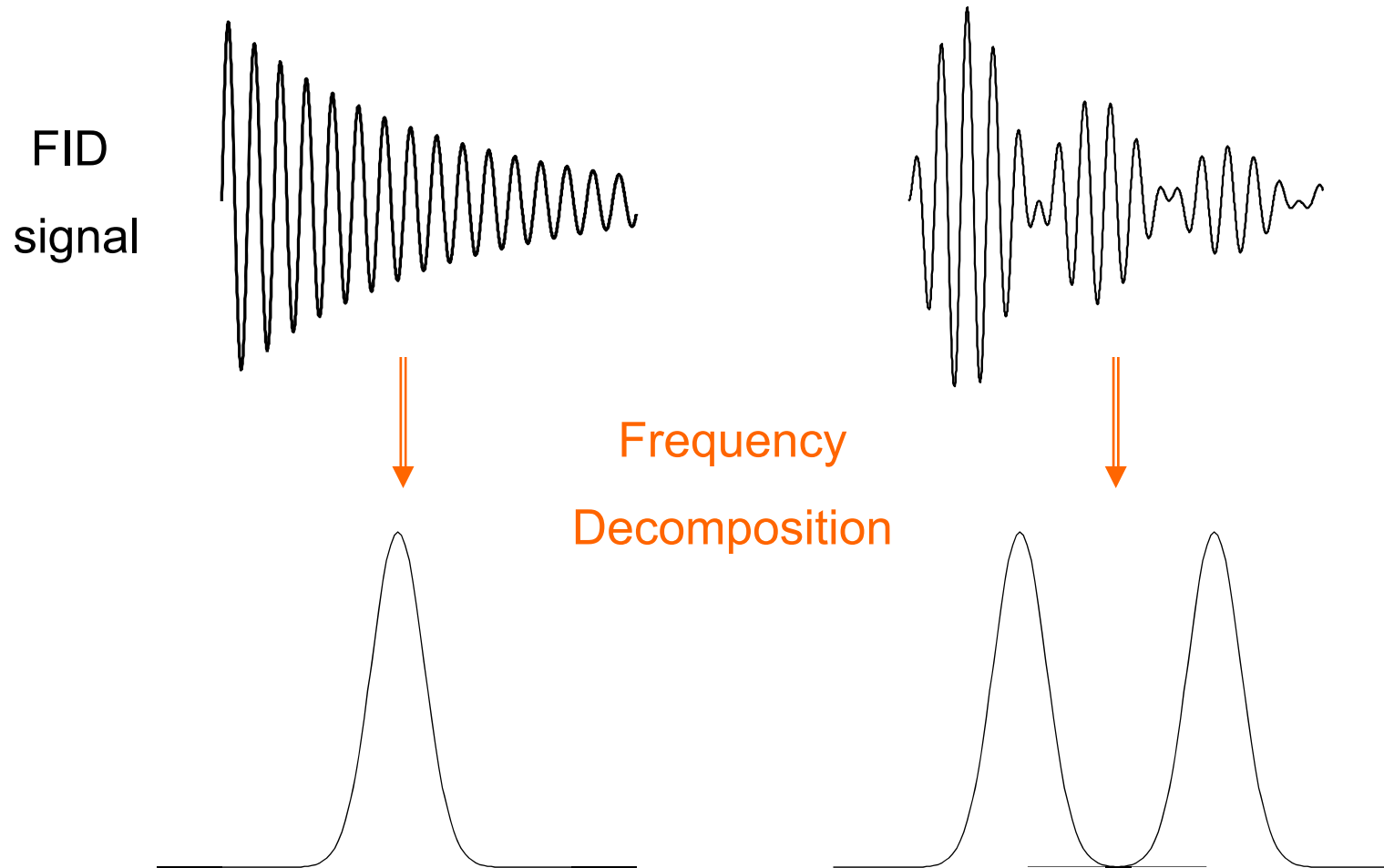
- During readout (image acquisition) period, a *frequency encoding* gradient field is applied --- using a deliberately applied non-uniform field to make the precession frequency depend on location
- Before readout (image acquisition) period, a *phase encoding* gradient field is applied --- during the readout (image acquisition) period. The effect of this gradient field is not time-varying but a fixed phase accumulation is determined by the amplitude and duration of the phase encoding gradient. $\Delta\Phi = \omega T_p$ where T_p is the time of Phase gradient application.



A Simple Example of Spatial Encoding



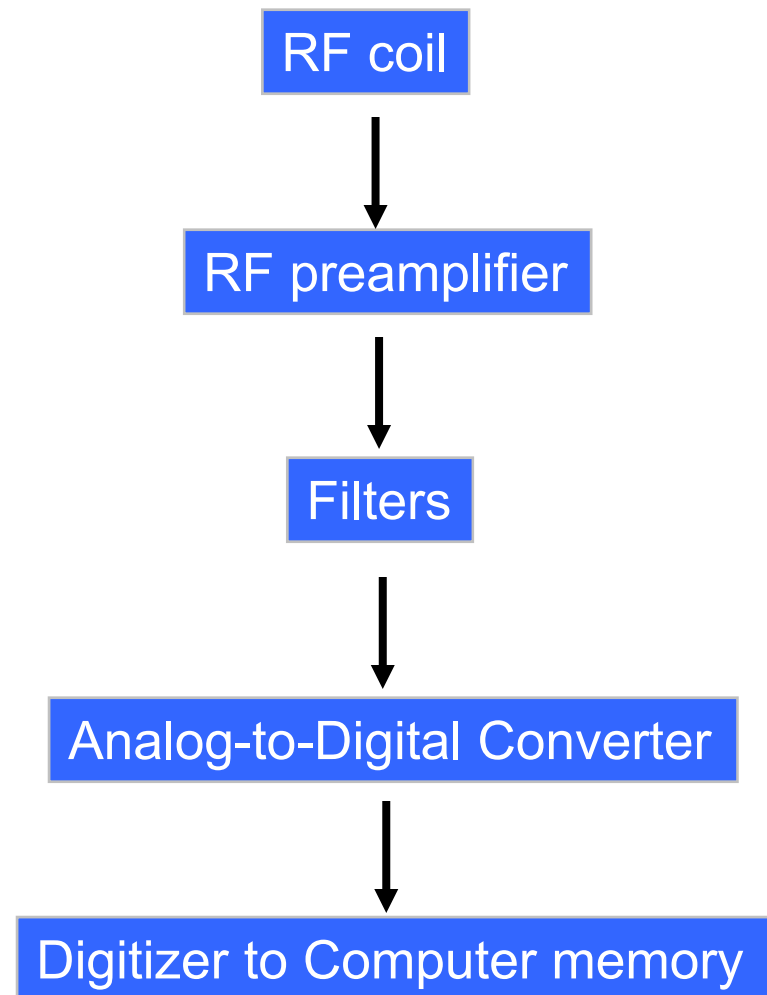
Spatial Decoding of the MR Signal



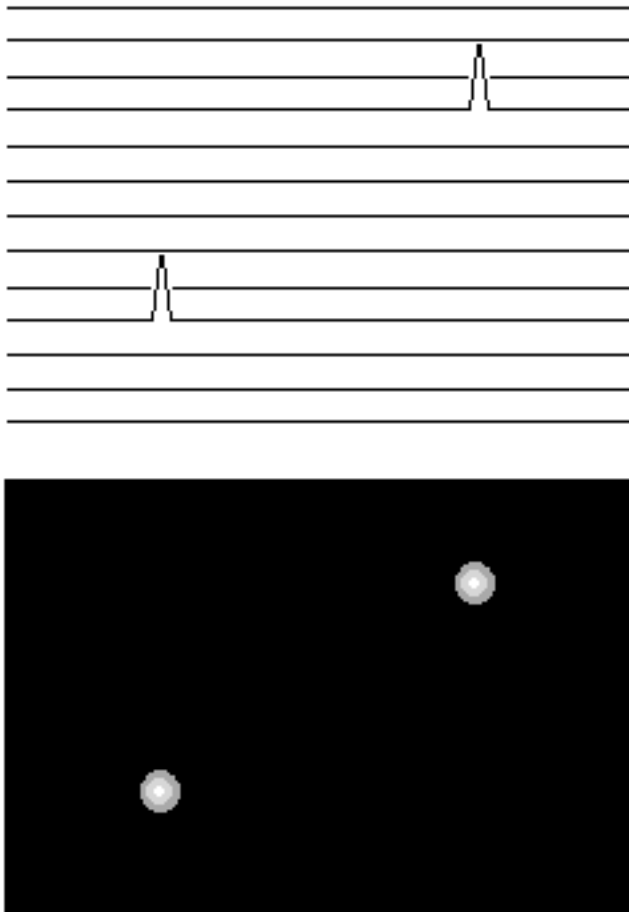
Receiving M_{xy} in Selected Z-Slice

Receive:

- During **readout**, gradient field perpendicular to slice selection gradient is turned on...
- Computer breaks measured signal $V(t)$ into frequency components $v(f)$ — using the Fourier transform
- Since frequency f varies across subject in a known way, we can assign each component $v(f)$ to the place it comes from



Final Image – FFT in Phase Direction



Lauterbur reconstructed the image by back-projecting the FT of k-space in several projections.

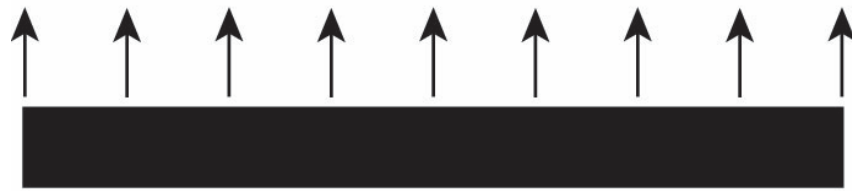
Usually the absolute value of the DFT of the raw data is taken as the pixel intensity value.

=> Raw k-space is richer than the final images we see!

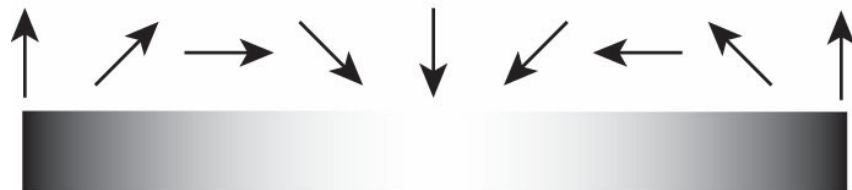


G_x applied

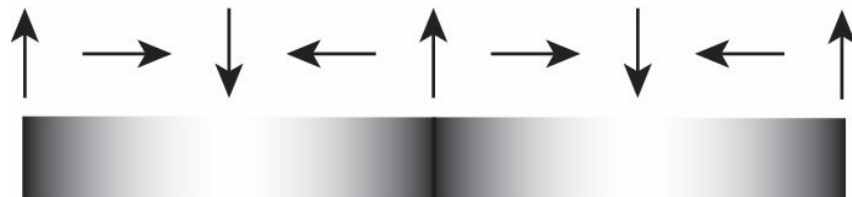
Arrows on right spin faster,
gaining phase over time



t_0



t_1



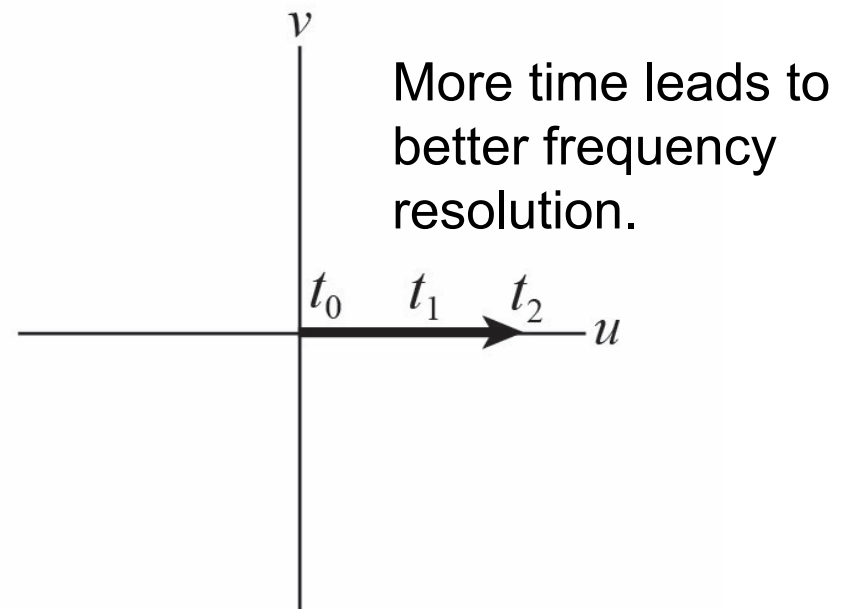
t_2

Arrows are $M_{x'y'}$ (in rotating frame)

The magnitude of each arrow is the
signal strength from that voxel.

k is spatial frequency

k -space traversed over time as
spatial frequency u increases



The total signal at a given time is a
particular spatial Fourier component.

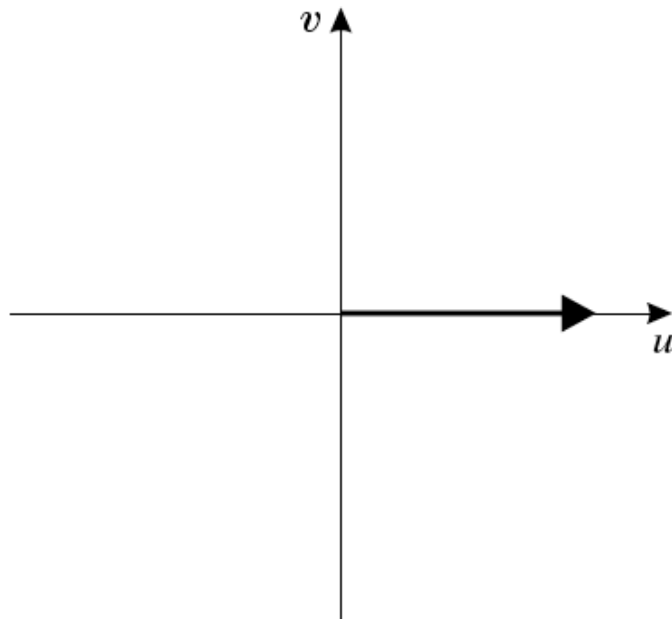
The temporal signal is complex. 72

k-space

- Arises from the convention in physics where wave number k represents a spatial frequency.
- Prince uses u and v instead of the traditional k_x and k_y

$$k_x = u \quad k_y = v$$

- Pulse sequences become a *Fourier trajectory* in *k-space*.



The single scan across the slice in the x direction just described is represented by this *Fourier trajectory*.

It is a trajectory in the sense that with passing time higher spatial frequencies are represented (larger phase angle for a given x).

$$\nearrow u = \gamma G_x t \nwarrow$$

spatial frequency is equivalent to temporal phase

If we can cover all of Fourier space, we can recover $f(x,y)$

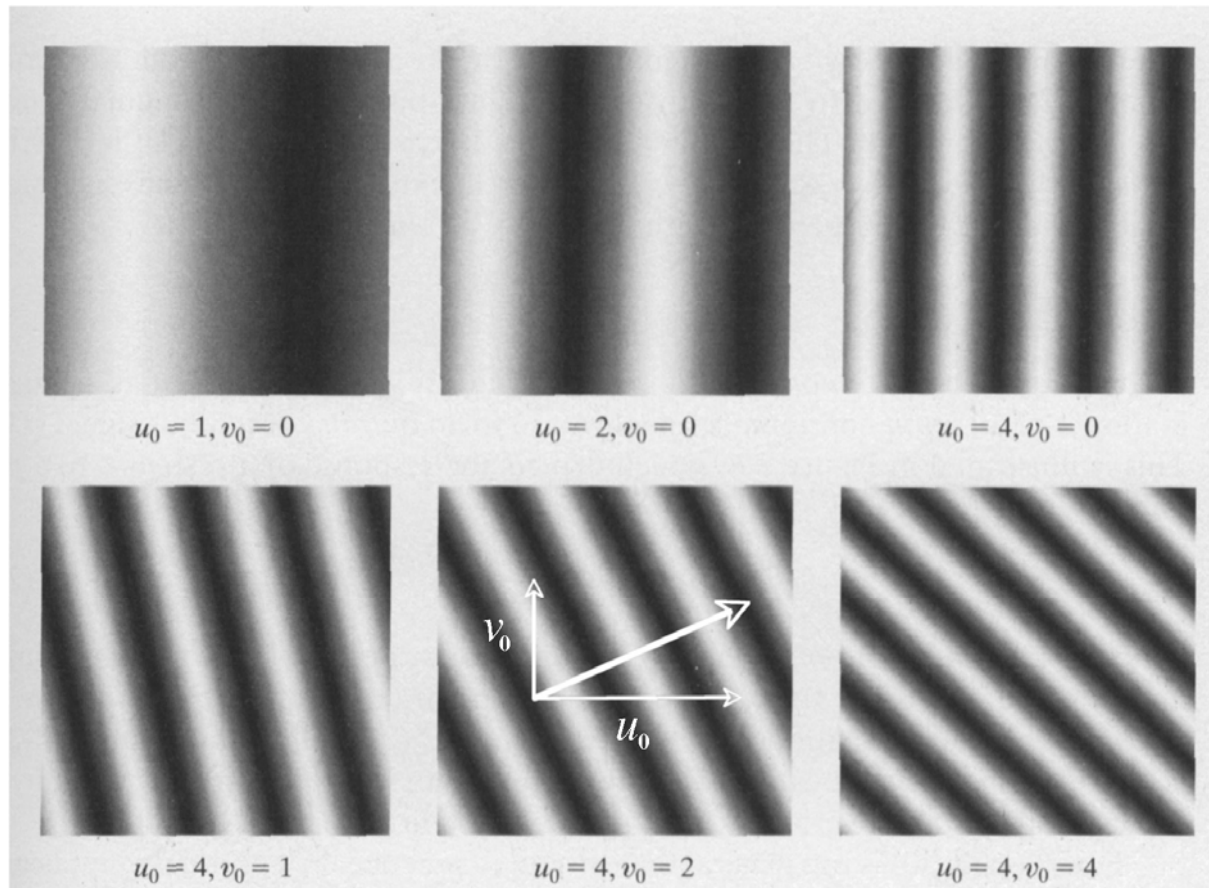
Recall Definition of Fourier in 2D

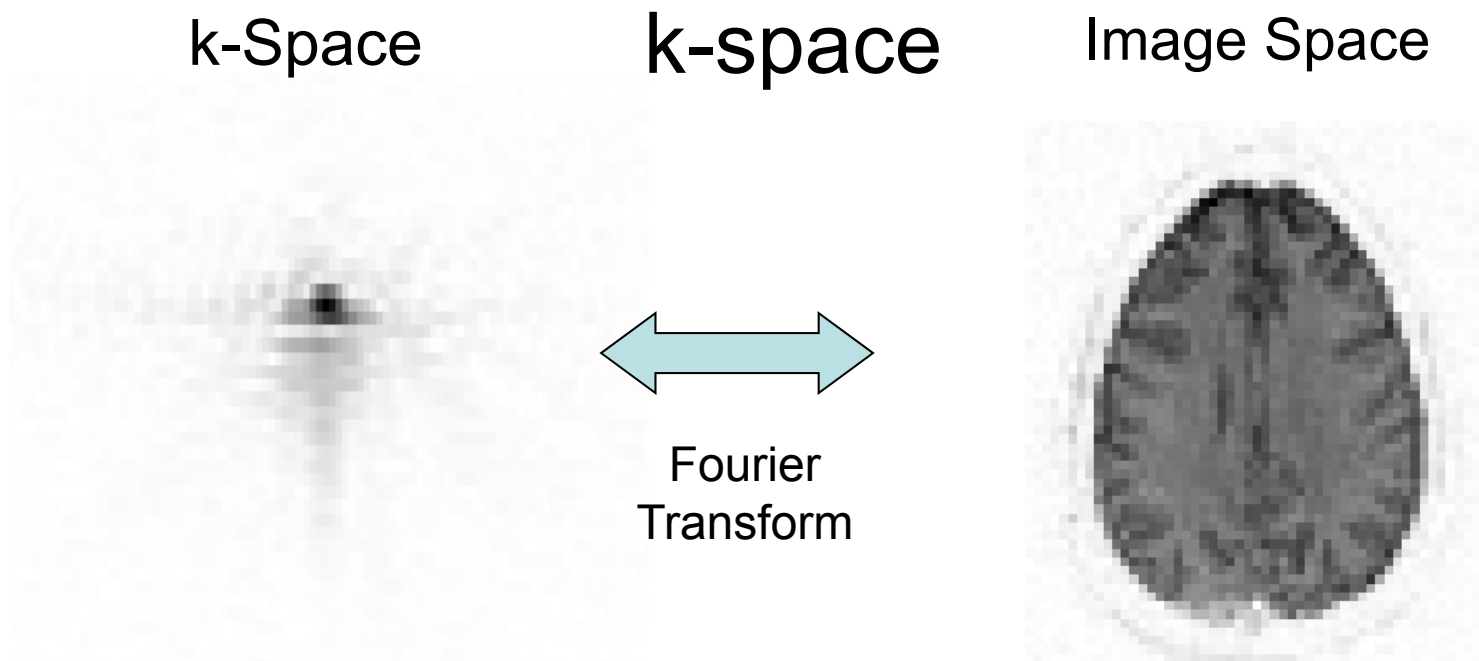
$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

in the rotating
frame of
reference

In k -Space
at any given
point of time,
local direction
of M_{xy} is the
phasor in the
above equations.
Spinning at
different rates at
different locations
they accumulate
different phase. 74





- k is the ‘wave number’, the inverse wavelength (number of crests per unit distance).
- Center of k-space represents lowest spatial frequencies (low frequencies contain “bulk” information of the image).
- Frequencies increase as you move from center (high frequencies contain details of image) .
- Spatial frequency in k-space corresponds to the relative phases of transverse magnetization accumulated by applying G_x (and/or G_y).

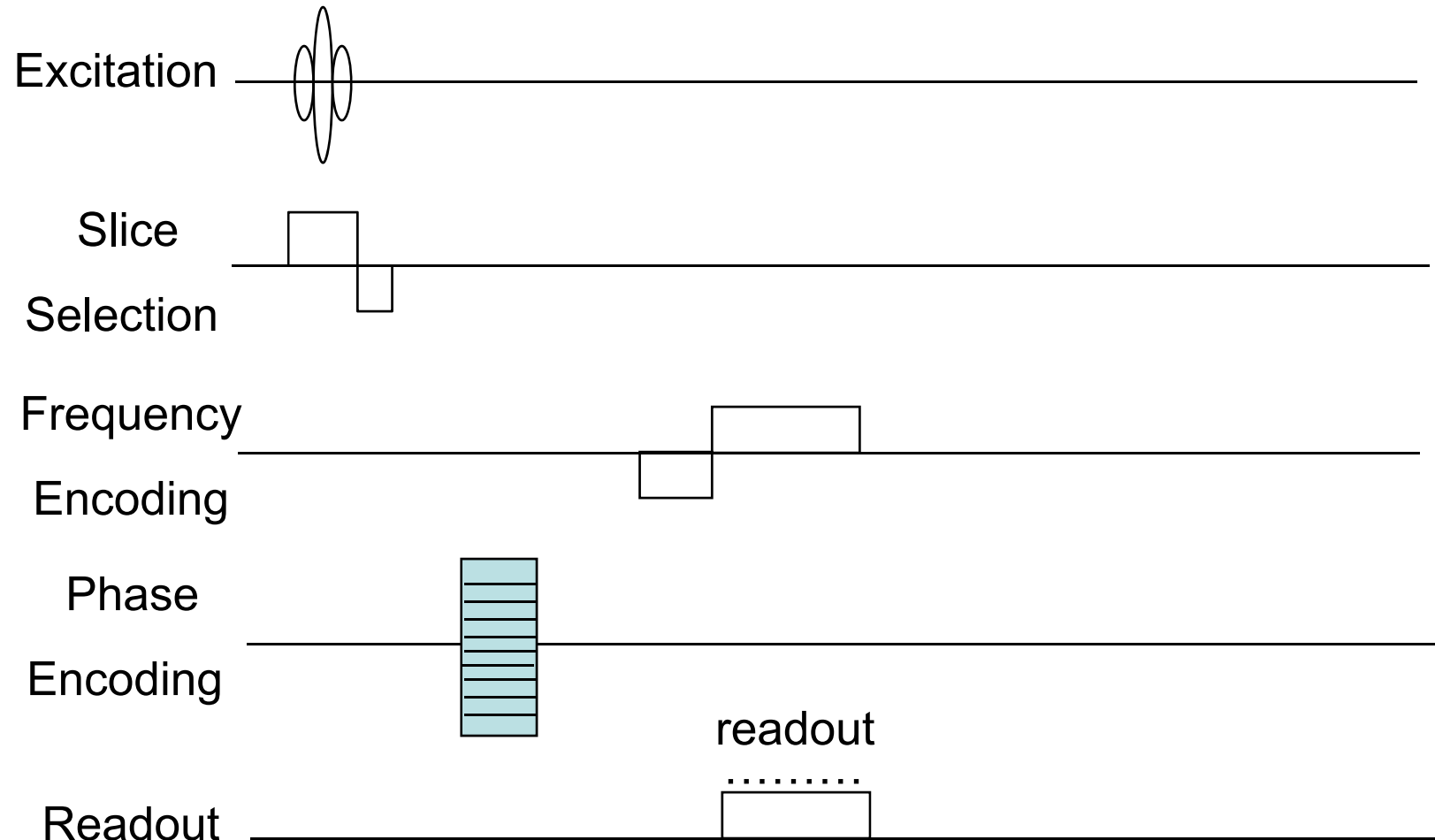
[Filling K-Space Animation](http://mrsrl.stanford.edu/~brian/mri-movies/k-space-outward.mov)

<http://mrsrl.stanford.edu/~brian/mri-movies/k-space-outward.mov>

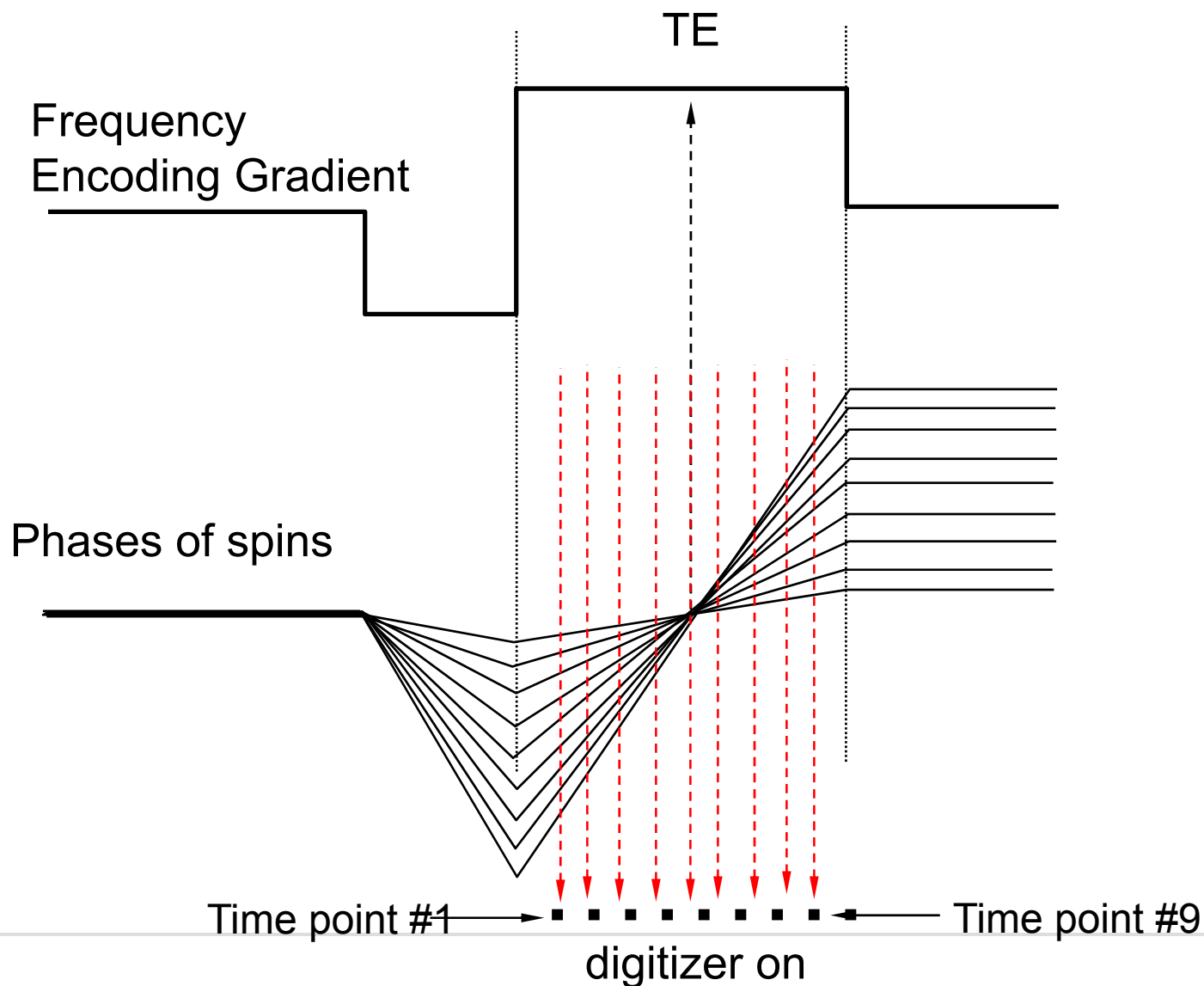
More advanced MRI Pulse Sequences

- Gradient Echo
- Spin Echo
- Fast Spin Echo
- Inversion Recovery

Pulse Sequence: Gradient-echo imaging

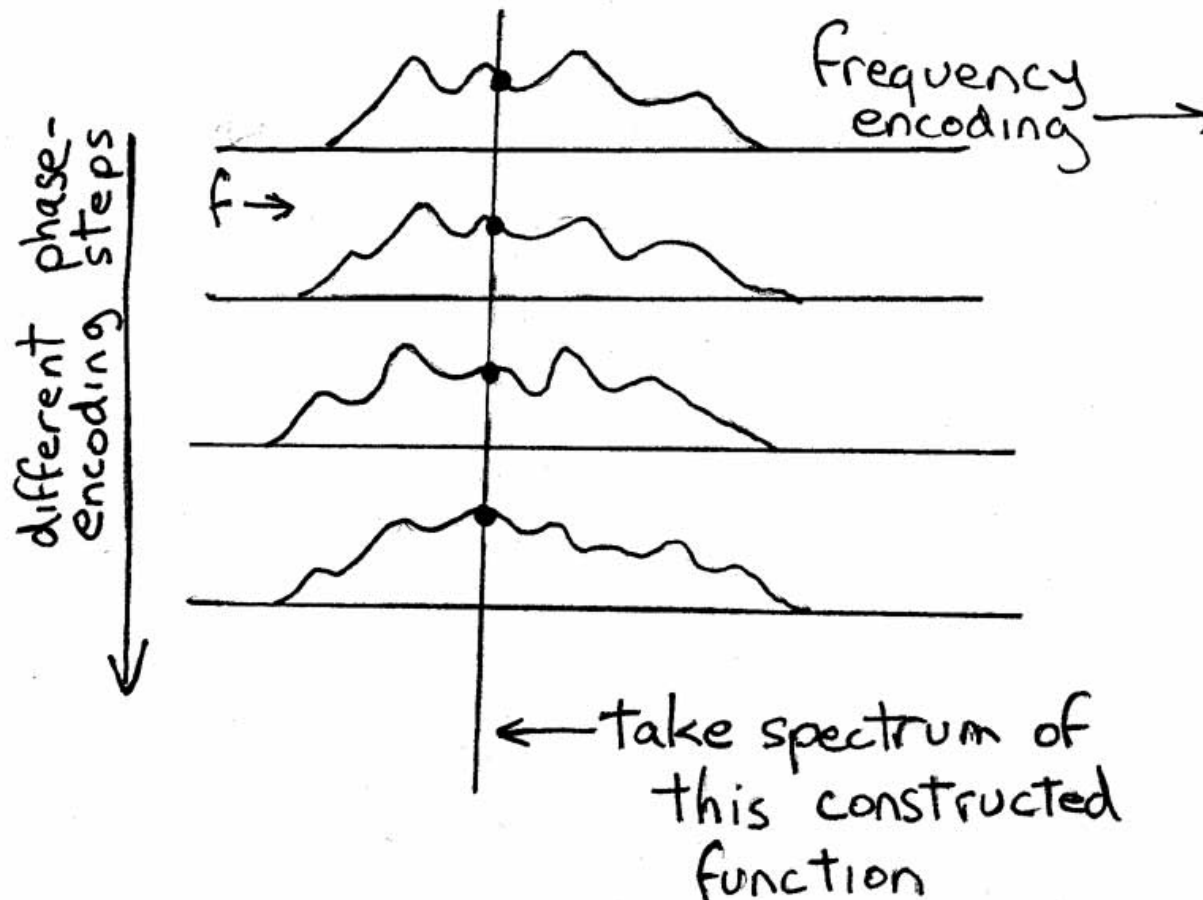


Phase Evolution of Gradient Echo MR Data with Frequency Encoding (i.e. w/o Phase Encoding)

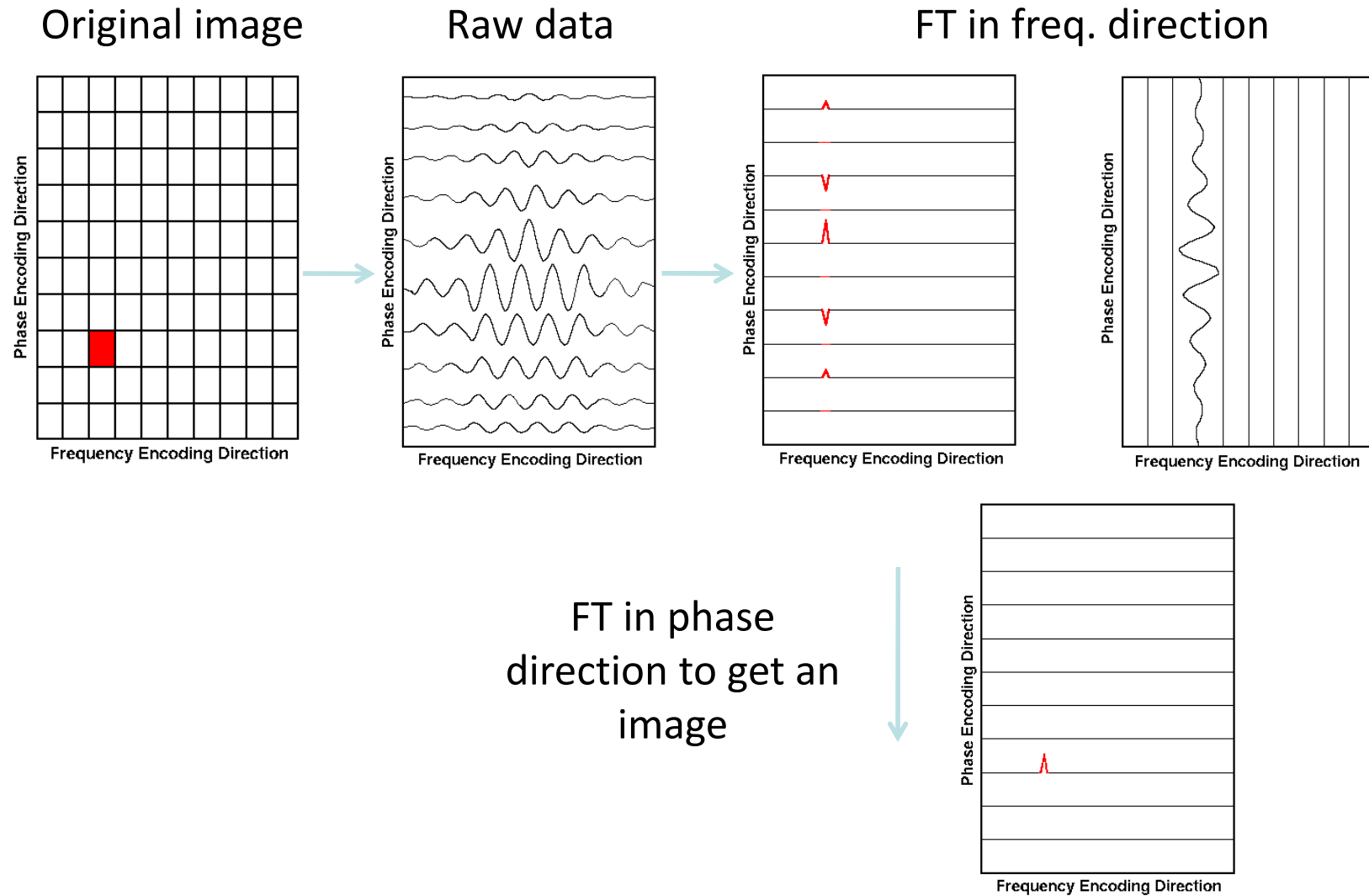


Phase Encoding

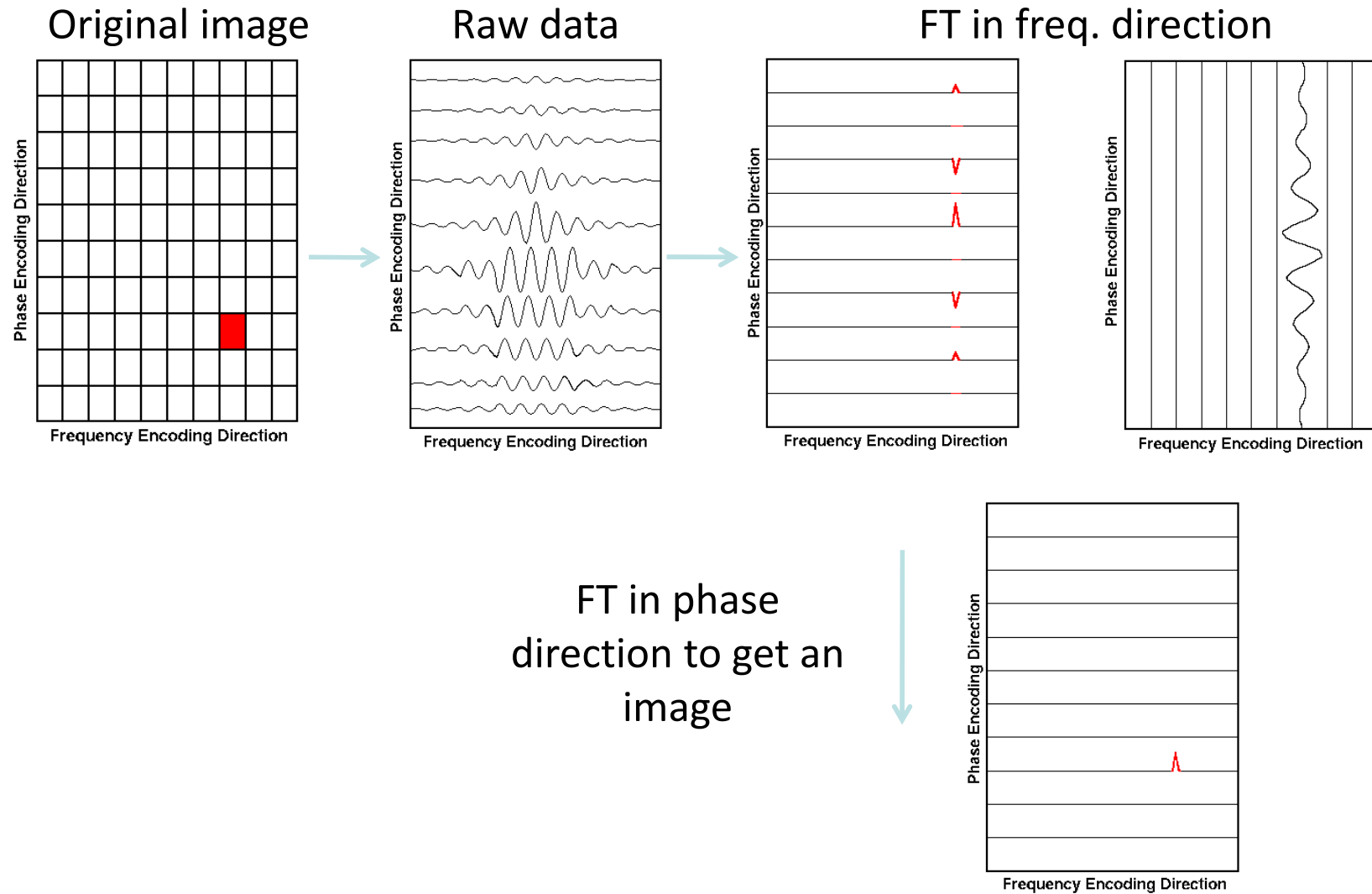
- Phase encoding looks at a particular read frequency across multiple acquisitions to construct a function sampled across the different phase-encoded signals. The FT of this “signal” yields “frequencies” each proportional to the phase increment at that point along the phase gradient G_y .



FID – Example 1



FID – Example 2



FID – Example 3

